List and Tuple – lab work

1. E-Commerce Order Analysis

Problem Statement

An online store records orders as:

```
orders = [
   ("Laptop", 55000),
   ("Mouse", 800),
   ("Keyboard", 1500),
   ("Monitor", 12000),
   ("Pen Drive", 600)
]
```

Write a program to:

- Display all products costing more than ₹1000.
- Find the most expensive product.
- Calculate the total order value.
- Count products costing below ₹1000.

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2. Cricket Tournament Scorecard

Problem Statement

A batsman's scores in different matches are stored in a list.

scores = [45, 78, 12, 100, 67, 8, 90, 55]

Write a program to:

- Count half-centuries and centuries.
- Find the highest score.
- Display all scores below 20.
- Calculate the average score.

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3. Smart Parking System

Problem Statement

Parking slots are represented as:

slots = [1, 0, 1, 1, 0, 0, 1, 0]

Where:

- 1 = Occupied
- 0 = Available

Write a program to:

- Count occupied and available slots.

- Find the first available slot.
- Display all available slot numbers.
- Check whether parking occupancy exceeds 75%.

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4. Employee Salary Processing

Problem Statement

Employee data is stored as tuples:

```
employees = [
  ("Rahul", 35000),
  ("Priya", 55000),
  ("Amit", 42000),
  ("Neha", 65000)
]
```

Write a program to:

- Display employees earning above ₹50,000.
- Find the highest-paid employee.
- Calculate total salary expenditure.
- Count employees earning below ₹40,000.

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5. Library Book Search

Problem Statement

Books available in a library:

```
books = [
  ("Python Basics", 5),
  ("Data Science", 0),
  ("Java Programming", 3),
  ("Machine Learning", 0)
]
```

Write a program to:

- Display unavailable books.
- Find all books with more than 2 copies.
- Count available books.
- Stop searching once a requested book is found.

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6. Student Attendance Tracker

Problem Statement

Attendance for 15 days is recorded as:

```
attendance = ['P', 'P', 'A', 'P', 'A', 'P', 'P', 'P', 'A', 'P', 'P', 'A', 'P', 'P', 'P']
```

Write a program to:

- Count present and absent days.
- Calculate attendance percentage.
- Determine eligibility (minimum 75% attendance).
- Display positions where the student was absent.

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7. Online Quiz Evaluation

Problem Statement

Correct answers:

correct = ['A', 'C', 'B', 'D', 'A']

Student answers:

student = ['A', 'B', 'B', 'D', 'C']

Write a program to:

- Calculate score.
- Display incorrectly answered question numbers.
- Count correct and wrong answers.
- Determine pass/fail (minimum 60%).

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8. Bus Route Monitoring

Problem Statement

Passenger count at each stop:

passengers = [12, 18, 25, 30, 28, 15, 8]

Write a program to:

- Find the busiest stop.
- Display stops with fewer than 10 passengers.
- Calculate average passengers.
- Determine whether any stop exceeded 25 passengers.

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9. Warehouse Product Inspection

Problem Statement

Product IDs and quality status:

```
products = [
  (101, "Pass"),
  (102, "Fail"),
  (103, "Pass"),
  (104, "Fail"),
```

```
  (105, "Pass")
]
```

Write a program to:

- Display failed product IDs.
- Count passed and failed products.
- Calculate pass percentage.
- Stop checking if 3 failures are found.

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10. Train Reservation Waiting List

Problem Statement

Passenger records:

```
passengers = [
  ("Anuj", "Confirmed"),
  ("Rahul", "Waiting"),
  ("Priya", "Confirmed"),
  ("Amit", "Waiting"),
  ("Neha", "Confirmed")
]
```

Write a program to:

- Display all waiting-list passengers.
- Count confirmed and waiting passengers.
- Find whether a specific passenger has a confirmed ticket.
- Create separate lists for confirmed and waiting passengers.